

proof $B_0^1(x) = 1$

$$\text{TGM: } \varphi(\lambda, \tau; x) = \frac{\tau^\lambda e^{\tau x}}{(e^\tau - 1)^\lambda} = \sum_{\nu=0}^{\infty} \frac{\tau^\nu}{\nu!} B_\nu^1(x)$$

Recuerdo $e^a = \sum_{x=0}^{\infty} \frac{a^x}{x!} = 1 + a + \frac{a^2}{2!} + \dots + \frac{a^n}{n!} + \dots$

Luego:

$$\frac{\tau^\lambda \sum_{r=0}^{\infty} \frac{(\tau x)^r}{r!}}{\left(\sum_{r=0}^{\infty} \frac{\tau^r}{r!} - 1 \right)^\lambda} = \sum_{\nu=0}^{\infty} \frac{\tau^\nu}{\nu!} B_\nu^1(x)$$

$$\tau^\lambda \left(1 + \tau x + \frac{\tau^2 x^2}{2!} + \dots \right) = \left[\cancel{1} + \tau + \frac{\tau^2}{2!} + \dots \cancel{-1} \right]^\lambda \sum_{\nu=0}^{\infty} \frac{\tau^\nu}{\nu!} B_\nu^1(x)$$

$$\tau^\lambda + \tau^{\lambda+1} x + \dots = \left[\tau^\lambda + \alpha \tau^{\lambda+1} + \dots \right] \left[B_0^1(x) + \frac{\tau}{1!} B_1^1(x) + \dots \right]$$

$$\tau^\lambda + \underbrace{\tau^{\lambda+1} x + \dots}_\theta = \tau^\lambda B_0^1(x) + \underbrace{\frac{\tau^{\lambda+1}}{1!} B_1^1(x) + \dots + \alpha \tau^{\lambda+1} B_0^1(x) + \dots}_\theta$$

$$\tau^\lambda + \theta = \tau^\lambda B_0^1(x) + \theta$$

Luego $\tau^\lambda = \tau^\lambda B_0^1(x)$ Entonces

$$B_0^1(x) = 1$$